

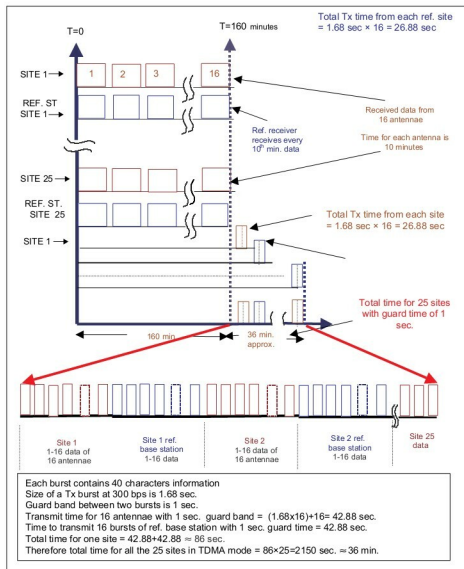


Fig. 7: Timing diagram of Rx data from antennae & Tx data burst to satellite

corrected and appended with unique word/bit time recovery/carrier recovery (see Fig. 5):

Size of data from each antenna = 40 characters (each character is 8 bytes long)

Identification bits of each antenna (ANT.ID) = 8 bits

Identification bits of each site (site ID) = 8 bits

Flag and dummy bits = 8 bytes + 8 bytes = 16 bits

CR = 128 bits, BTR = 64 bits and UW = 64 bits

Therefore the total number of bits after scrambling + 8 dummy bits = 376 bits

After FEC coding 1008 symbols:

Burst time after FEC coding = 1.68 sec.

onds at 600 symbols/second

Site ID may not be required if separate carrier is used for each site.

Space segment. Some of the INSAT series satellites have a UHF transponder specifically designed for meteorological data collection applications. The frequency band available for this application is 402.65 to 402.85 MHz.

Presently, two Indian geo-synchronous satellites are operational, having data relay transponders at KALPANA satellite at location 74°E and INSAT 3A at 93°E. Satellite parameters and satellite link calculation are shown in Tables II and III.

Total C/No = 39.5 dBHz
Required C/No = 34 dBHz
(for Eb/No = 9 dB)

Therefore

Link margin = Total C/No -

Required C/No

= 5.5 dB

From the above calculation, the required UHF transmit power, for transmit data rate of 300 bps (1/2 rate), is maximum 10W, which is sufficient to receive at the hub station with a 3.8m antenna (G/T of 17 dB/K) with sufficient link margin of 5.5 dB.

Receive station (hub). The receive system is shown in Fig. 6. It consists of a 3.8m antenna, RF down-converter, IF down-converter, BPSK burst demodulator and data processing system (computer).

Antenna system. The satellite signal in the Ext.C band is received by the 3.8m antenna of the ground terminal. This antenna is a solid dish type with G/T of 17 dB/K. It is adjustable over 0-90° in elevation and 360° in azimuth axes.

RF system. The RF system consists of a low-noise amplifier (LNA) and an RF down-converter. The low-noise block (LNB) down-converter fitted with the feed system amplifies the low-level signal received by the antenna. The output of the LNB goes to the L-band

down-converter, which converts the RF signal into 70MHz IF signal. The IF signal is in the 70±20MHz band. It is fed to the IF down-converter and the DSP-based burst demodulator. The demodulator output at 300 bps is fed to the computer through a synchronous hardware unit.

Synchronous hardware unit (SHU). The SHU receives the data packet from the demodulator and calculates CRC of the data packet. If CRC is correct, the packet undergoes further processing. Else, it is discarded. The repeated data packet is also discarded. The data is decompressed and reformatted to the original form and stored in the file to